

5 月 月考试卷 (题目)

考试时间: 3 小时 总分: 150 分

本次月考出题范围: **整个高数上册**

一、选择题 (每小题 5 分, 共 50 分)

1. 设当 $x \rightarrow 0$ 时, $\int_0^x \frac{(1+e^{t^2})\sin t^2}{\ln(t+\sqrt{1+t^2})} dt$ 为 x^k 的同阶无穷小量, 则 $k = (\quad)$

A. 1 B. 2 C. 3 D. 4

解: $f(x) \sim \int_0^x \frac{2t^2}{t} dt = \int_0^x 2t dt = x^2$, 故 $k=2$, 选 B.

2. 设 $f(x) = \frac{x}{\cos x}$ 在 $x=0$ 处的三次泰勒多项式为 $ax + bx^2 + cx^3$, 则 $(\quad)$

A. $a=1, b=0, c=\frac{1}{2}$

B. $a=0, b=1, c=\frac{1}{2}$

C. $a=0, b=\frac{1}{2}, c=1$

D. $a=\frac{1}{2}, b=1, c=0$

解:

$$b=0 \Rightarrow f(x) \cdot \cos x \Rightarrow (ax + cx^3) \cdot \left(1 - \frac{x^2}{2} + \dots\right) = x$$

$$\Rightarrow ax - \frac{a}{2}x^3 + cx^3 = x \Rightarrow \begin{cases} a=1 \\ c=\frac{1}{2} \end{cases}$$

故选 A.

3. 设常数 $a > 0$, 当 $x \in (1, +\infty)$ 时, $\ln x \leq x^a$, 则 a 的范围为 $(\quad)$

A. $(0, e]$

B. $\left(0, \frac{1}{e}\right)$

C. $[e, +\infty)$

D. $\left[\frac{1}{e}, +\infty\right)$

$$\ln \ln x \leq a \cdot \ln x \Rightarrow a \geq \frac{\ln \ln x}{\ln x} \xrightarrow{\ln x=t} \frac{\ln t}{t} = f(t) \quad (t > 0)$$

解: $f'(t) = \frac{1-\ln t}{t^2}$. $f'(e) = 0 \Rightarrow f(t)$ 在 $(0, e)$ 递增, 在 $(e, +\infty)$ 递减

$$\Rightarrow [f(x)]_{\max} = f(e) = \frac{1}{e}$$

故选 D.



4. 设 $f(x) = \begin{cases} x, & x < 0 \\ e^x, & x \geq 0 \end{cases}$, $g(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, 则 ()

A. $f(x)$ 在 $[-1, 1]$ 上有原函数、可积

B. $\int_{-1}^x f(t) dt$ 在 $[-1, 1]$ 上不连续

C. $g(x)$ 在 $[-1, 1]$ 上无原函数、不可积

D. $G(x) = \int_{-1}^x g(t) dt$ 在 $[-1, 1]$ 可导, 且 $G'(x) = g(x)$

解: 选 D. (看凯哥 b 站视频: 变限积分是否可导, 记住结论可以直接用)

选项 A: 存在第一类间断点, 所以不存在原函数;

选项 B: 设 $F(x) = \int_{-1}^x f(t) dt$, $\lim_{x \rightarrow 0^-} F(x) = -\frac{1}{2}$, $\lim_{x \rightarrow 0^+} F(x) = -\frac{1}{2} = F(0)$,

故 $F(x)$ 在 $[-1, 1]$ 上连续;

选项 C: 可积且存在原函数;

选项 D: 由微积分基本定理知 $G(x) = \int_{-1}^x g(t) dt$ 在 $[-1, 1]$ 可导, 且 $G'(x) = g(x)$

5. 若反常积分 $\int_0^{+\infty} \frac{\arctan^p x}{\sqrt{x}(1+x^{2-p})} dx$ 收敛, 则 p 的取值范围为 ()

A. $-\frac{1}{2} < p < \frac{1}{2}$

B. $-\frac{1}{2} < p < \frac{3}{2}$

C. $\frac{1}{2} < p < \frac{3}{2}$

D. $\frac{3}{2} < p < \frac{5}{2}$

解: $x \rightarrow +\infty, f(x) \sim \left(\frac{\pi}{2}\right)^p \cdot \frac{1}{\sqrt{x} \cdot (1+x^{2-p})} \sim \left(\frac{\pi}{2}\right)^p \cdot \frac{1}{x^{\frac{5}{2}-p}}$ 在 $\frac{5}{2} - p > 1$ 时收敛 $\Rightarrow p < \frac{3}{2}$

$x \rightarrow 0^+, f(x) \sim \frac{x^p}{\sqrt{x} \cdot (1+x^{2-p})} \sim \frac{x^p}{\sqrt{x}} = \frac{1}{x^{\frac{1}{2}-p}}$ 在 $\frac{1}{2} - p < 1$ 时收敛 $\Rightarrow p > -\frac{1}{2}$

综上 $-\frac{1}{2} < p < \frac{3}{2}$, 选 B.

6. 设 $f(x) = (x-1)(x-2)^2(x-3)^3 \cdots (x-n)^n (n \geq 3)$, 则 $f^{(n)}(n) = ()$

A. $n \cdot n!$

B. $(n!)^n$

C. $(n!)!$

D. $y = n! \cdot (n-1)! \cdots 3! \cdot 2! \cdot 1!$

解:

$$\lim_{x \rightarrow n} \frac{f(x)}{(x-n)^n} = \lim_{x \rightarrow n} (x-1)(x-2)^2 \cdots [x-(n-1)]^{n-1} = (n-1)(n-2)^2(n-3)^3 \cdots 1^{n-1} = (n-1) \cdots 1^{n-1}$$

$$\Rightarrow \lim_{x \rightarrow n} \frac{f(n) + f'(n)(x-n) + \cdots + \frac{f^{(n-1)}(n)}{(n-1)!}(x-n)^{n-1} + \frac{f^{(n)}(n)}{n!}(x-n)^n + o[(x-n)^n]}{(x-n)^n}$$

$$\Rightarrow f(n) = f'(n) = \cdots = 0, f^{(n-1)}(n) = 0$$

$$\frac{f^{(n)}(n)}{n!} = (n-1)(n-2)^2 \cdots 1^{n-1} \Rightarrow f^{(n)}(n) = n!(n-1)(n-2)^2(n-3)^3 \cdots 1^{n-1}$$

$$= n!(n-1)!(n-2)!(n-3)! \cdots 1$$

7. 设正值函数 $f(x)$ 二阶可导, $f(x)f''(x) > [f'(x)]^2$, 且 $f(0)=1, f'(0)=1$, 则()

- A. $e^{-x}f(x) \geq 1$ B. $e^{-x}f(x) < 1$ C. $xf(x) \geq 1$ D. $xf(x) < 1$

解:

$$f \cdot f'' - [f']^2 > 0 \Rightarrow \frac{f \cdot f'' - (f')^2}{f^2} > 0 \Leftrightarrow \left(\frac{f'}{f}\right)' = \frac{f'' \cdot f - (f')^2}{f^2} > 0$$

$$\Leftrightarrow [\ln f(x)]' > 0, \text{ 令 } g(x) = \ln f(x) \Rightarrow g'(x) > 0 \Rightarrow g(x) \geq g(0) + g'(0)x \Rightarrow \ln f(x) \geq \ln f(0) + \frac{f'(0)}{f(0)} \cdot x$$

$$\Rightarrow \ln f(x) \geq 0 + x \Rightarrow f(x) \geq e^x \Rightarrow e^{-x} \cdot f(x) \geq 1$$

故选 A.

8. 设 $g(x)$ 连续, $f(x)$ 连续可导, $\lim_{x \rightarrow 0} \frac{\ln[1+g(x)]}{x} = 2$, $f'(x) = -2\ln(1+x^2) + x \int_0^1 g(xt) dt$, 则()

- A. 点 $x=0$ 为 $f(x)$ 的极小值点
B. 点 $x=0$ 为 $f(x)$ 的极大值点
C. 点 $(0, f(0))$ 为曲线 $y=f(x)$ 的拐点
D. 点 $x=0$ 不是 $f(x)$ 的极值点, 点 $(0, f(0))$ 也不是曲线 $y=f(x)$ 的拐点

解: 易得

$$\lim_{x \rightarrow 0} g(x) = 0 \Rightarrow \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0} \Rightarrow g'(0) = 2,$$

$$f'(x) = -2\ln(1+x^2) + \int_0^1 g(xt) d(xt) = -2\ln(1+x^2) + \int_0^x g(u) du$$

$$f'(0) = 0, f''(x) = -2 \cdot \frac{2x}{1+x^2} + g(x) \Rightarrow f''(0) = 0 + g(0) = 0$$

$$f'''(x) = (-4) \cdot \frac{(1+x^2) - x \cdot 2x}{(1+x^2)^2} + g'(x) \Rightarrow f'''(0) = -4 \cdot \frac{1}{1} + g'(0) = -2 \neq 0$$

故选 C.

9. 设 $f(x)$ 在 $(0, +\infty)$ 内二阶可导, 则()

- A. 若 $\lim_{x \rightarrow +\infty} f(x) = 0$, 则 $\lim_{x \rightarrow +\infty} f'(x) = 0$
B. 若 $\lim_{x \rightarrow +\infty} f'(x) = 0$, 则 $\lim_{x \rightarrow +\infty} f(x) = 0$
C. 若 $\lim_{x \rightarrow +\infty} f(x) = +\infty$, 则 $\lim_{x \rightarrow +\infty} f''(x) = 0$
D. 若 $\lim_{x \rightarrow +\infty} f''(x) > 0$, 则 $\lim_{x \rightarrow +\infty} f(x) = +\infty$

解: 选项 A: 反例: $f(x) = \frac{\sin x^2}{x}$; 选项 B: 反例: $f(x) \equiv 1$;

选项 C: 反例: $f(x) = e^x$; 选项 D: $\lim_{x \rightarrow +\infty} f''(x) = k, \text{ 且 } k > 0$, 故选 D.



10. 设 $f(x)$ 非负连续, $f(x) \int_0^x f(x-t)dt = \sin^4 x$, 则 $f(x)$ 在 $\left[0, \frac{\pi}{2}\right]$ 上的平均值为 ()

- A. $\sqrt{\frac{3}{2\pi}}$ B. $\sqrt{\frac{3}{\pi}}$ C. $\sqrt{\frac{2}{3\pi}}$ D. $\sqrt{\frac{1}{\pi}}$

解: 选 A.

$$\begin{aligned} \bar{f} &= \frac{\int_0^{\frac{\pi}{2}} f(x) dx}{\frac{\pi}{2} - 0} \Rightarrow f(x) \cdot \int_0^x f(t) dt = \sin^4 x, \text{ 令 } F(x) = \int_0^x f(t) dt \Rightarrow 2F'(x) \cdot F(x) = \sin^4 x \\ &\Rightarrow F^2(x) = 2 \cdot \int_0^x \sin^4 t dt. \text{ 令 } x = \frac{\pi}{2} \Rightarrow \left[F\left(\frac{\pi}{2}\right)\right]^2 = 2 \cdot \int_0^{\frac{\pi}{2}} \sin^4 t dt. \\ &= 2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{8} \Rightarrow F\left(\frac{\pi}{2}\right) = \sqrt{\frac{3\pi}{8}} \Rightarrow \bar{f} = \frac{2}{\pi} \sqrt{\frac{3\pi}{8}} = \sqrt{\frac{3}{2\pi}} \end{aligned}$$

二、填空题 (每小题 5 分, 共 30 分)

11. $\lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos(\tan x)}{(e^x - 1)(\arcsin x - x)} = \underline{\hspace{2cm}}.$

解:

$$I = \lim_{x \rightarrow 0} \frac{(\sin x - \tan x) \cdot (-\sin \xi)}{x \cdot \frac{1}{6} x^3}, \text{ 其中 } \xi \text{ 介于 } \sin x \text{ 和 } \tan x \text{ 之间.}$$

由于 $x \rightarrow 0$ 时, $\sin x \sim \tan x \sim x$, 故 $x \rightarrow 0$ 时, $\xi \sim x$.

$$\Rightarrow I = \lim_{x \rightarrow 0} \frac{\left(-\frac{1}{2}x^3\right) \cdot (-x)}{x \cdot \frac{1}{6}x^3} = \frac{\frac{1}{2}}{\frac{1}{6}} = 3$$

12. $\lim_{n \rightarrow \infty} \sum_{i=n+1}^{3n} \frac{n}{i(n+i)} = \underline{\hspace{2cm}}.$

解:

$$\begin{aligned} I &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=n+1}^{3n} \frac{1}{\frac{i}{n} \cdot \left(1 + \frac{i}{n}\right)} = \int_1^3 \frac{1}{x(x+1)} dx \\ &= \int_1^3 \left(\frac{1}{x} - \frac{1}{x+1}\right) dx = \ln \frac{x}{x+1} \Big|_1^3 = \ln \frac{3}{\frac{1}{2}} = \ln \frac{3}{2} \end{aligned}$$

13. $\int_{-\pi}^{\pi} \frac{|x| \sin^2 x}{1 + e^{-2027x}} dx = \underline{\hspace{2cm}}.$

解:

$$\begin{aligned} I &= \int_{-\pi}^{\pi} \frac{|x| \cdot \sin^2 x}{1 + e^{2027x}} dx = \frac{1}{2} \int_{-\pi}^{\pi} |x| \cdot \sin^2 x \cdot \left(\frac{1}{1 + e^{-2027x}} + \frac{1}{1 + e^{2027x}}\right) dx \\ &= \frac{1}{2} \int_{-\pi}^{\pi} |x| \cdot \sin^2 x dx = \int_0^{\pi} x \cdot \sin^2 x dx = \pi \cdot \int_0^{\frac{\pi}{2}} \sin^2 x dx = \pi \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi^2}{4} \end{aligned}$$

14. $\int \left(\frac{x-1}{x^2+1}\right)^2 e^x dx = \underline{\hspace{2cm}}.$

解:

$$I = \int e^x \cdot \frac{(x^2+1)-2x}{(x^2+1)^2} dx = \int e^x \cdot \left[\frac{1}{x^2+1} + \frac{-2x}{(x^2+1)^2} \right] dx$$

$$\text{注意到: } \left(\frac{1}{x^2+1} \right)' = \frac{-2x}{(x^2+1)^2} \Rightarrow I = e^x \cdot \frac{1}{x^2+1} + C$$

15. 微分方程 $y' = \tan y + xe^x \sec y$ 的通解为 _____.

解:

$$\cos y \cdot y' = \sin y + xe^x \Rightarrow \cos y \cdot y' - \sin y = xe^x$$

$$\Rightarrow e^{-x} \cdot [\cos y \cdot y' - \sin y] = x \Rightarrow (e^{-x} \sin y)' = x \Rightarrow e^{-x} \cdot \sin y = \frac{1}{2} x^2 + C \Rightarrow \sin y = e^x \left(\frac{1}{2} x^2 + C \right)$$

16. 设 $f(x) = x^2 3^x$, 则 $f^{(100)}(1) = \underline{\hspace{2cm}}$.

解:

$$\begin{aligned} f^{(100)}(x) &= C_{100}^0 \cdot x^2 \cdot (3^x)^{(100)} + C_{100}^1 \cdot 2x \cdot (3^x)^{(99)} + C_{100}^2 \cdot 2 \cdot (3^x)^{(98)} \\ &= 1 \cdot x^2 \cdot 3^x \cdot (\ln 3)^{100} + 200 \cdot x \cdot 3^x \cdot (\ln 3)^{99} + 100 \cdot 99 \cdot 3^x \cdot (\ln 3)^{98} \\ \text{令 } x=1 &\Rightarrow f^{(100)}(1) = 3 \cdot (\ln 3)^{100} + 600 \cdot (\ln 3)^{99} + 29700 \cdot (\ln 3)^{98} \end{aligned}$$

三、解答题 (共 6 题, 总共 70 分)

17. (本题 10 分)

$$\text{设 } f(x) \text{ 可导, 且 } f(x) = x + x \int_0^1 f(x) dx + x^2 \lim_{x \rightarrow 0} \frac{f(x)}{x}, \text{ 求 } f(x).$$

解:

$$\text{设 } \int_0^1 f(x) dx = A, \lim_{x \rightarrow 0} \frac{f(x)}{x} = B \quad (*)$$

$$\text{在 } (*) \text{ 积分: } A = \frac{1}{2} + \frac{1}{2} A + \frac{1}{3} B$$

$$\text{在 } (*) \text{ 式两边除以 } x, \text{ 并令 } x \rightarrow 0 \Rightarrow B = 1 + A + \lim_{x \rightarrow 0} x \cdot B$$

$$\Rightarrow A = \frac{1}{2} + \frac{1}{2} A + \frac{1}{3} + \frac{1}{3} A \Rightarrow \frac{1}{6} A = \frac{5}{6} \Rightarrow A = 5 \Rightarrow B = 1 + 5 = 6$$

$$\Rightarrow f(x) = x + 5x + 6x^2 \Rightarrow f(x) = 6x^2 + 6x$$

18. (本题 10 分)

$$(1) \text{ 若 } x \rightarrow 0 \text{ 时, } (1+x)^{\frac{1}{x}} = a + bx + cx^2 + o(x^2), \text{ 求 } a, b, c;$$

$$(2) \lim_{x \rightarrow +\infty} x^2 \left[\left(1 + \frac{1}{x+1} \right)^{x+1} - \left(1 + \frac{1}{x} \right)^x \right]$$

解: (1)



$$f(x) = e^{\frac{\ln(1+x)}{x}} = e^1 \cdot e^{\frac{\ln(1+x)-x}{x}}$$

$$\Rightarrow \frac{\ln(1+x)-x}{x} = -\frac{x}{2} + \frac{x^2}{3} - \frac{x^3}{4} + \dots \sim -\frac{x}{2}$$

$$\text{当 } x \rightarrow 0 \text{ 时, } f(x) = e \cdot \left[1 + \frac{\ln(1+x)-x}{x} + \frac{1}{2!} \cdot \left(\frac{\ln(1+x)-x}{x} \right)^2 + o\left(\left(\frac{\ln(1+x)-x}{x} \right)^2 \right) \right]$$

$$= e \cdot \left[1 + \left(-\frac{x}{2} + \frac{x^2}{3} \right) + \frac{1}{2!} \cdot \left(-\frac{x}{2} \right)^2 + o(x^2) \right]$$

$$= e \cdot \left[1 - \frac{x}{2} + \frac{11}{24}x^2 + o(x^2) \right]$$

$$\text{故 } a = e, b = -\frac{e}{2}, c = \frac{11e}{24}$$

(2)

$$I = \lim_{x \rightarrow +\infty} x^2 \cdot \left[e \cdot \left(1 - \frac{1}{2} \cdot \frac{1}{x+1} + \frac{11}{24} \cdot \frac{1}{(x+1)^2} \right) - e \cdot \left(1 - \frac{1}{2} \cdot \frac{1}{x} + \frac{11}{24} \cdot \frac{1}{x^2} \right) + o\left(\frac{1}{x^2} \right) \right]$$

$$= e \cdot \lim_{x \rightarrow +\infty} x^2 \left[\frac{1}{2} \left(\frac{1}{x} - \frac{1}{x+1} \right) + \frac{11}{24} \left(\frac{1}{x^2} - \frac{1}{(x+1)^2} \right) + o\left(\frac{1}{x^2} \right) \right]$$

$$= e \cdot \lim_{x \rightarrow +\infty} x^2 \cdot \frac{1}{2} \cdot \frac{1}{x(x+1)}$$

$$= \frac{e}{2}$$

19. (本题 12 分)

$$\text{设 } f(x) = x - [x], \text{ 求 } \lim_{x \rightarrow +\infty} \frac{\int_0^x f(t) |\sin \pi t| dt}{x}.$$

解:

$$g(x) = |\sin \pi x| \Rightarrow f(x) \text{ 和 } g(x) \text{ 周期均为 } 1$$

$$\text{令 } S(x) = \int_0^x f(t) \cdot |\sin \pi t| dt, S'(x) = f(x) \cdot |\sin \pi x| \geq 0 \text{ 且等号仅在 } x = 1, 2, 3, \dots \text{ 时取到}$$

$$\Rightarrow S(x) \text{ 严格递增, 当 } n \leq x < n+1 \text{ 时 } \Rightarrow S(n) \leq S(x) < S(n+1)$$

其中

$$S(n) = \int_0^n f(t) \cdot |\sin \pi t| dt = n \int_0^1 f(t - [t]) \cdot |\sin \pi t| dt$$

$$= n \int_0^1 t \cdot \sin \pi t dt$$

$$= \left(-\frac{n}{\pi} \right) \int_0^1 t d(\cos \pi t)$$

$$= \left(-\frac{n}{\pi} \right) \cdot \left[t \cdot \cos \pi t \Big|_0^1 - \int_0^1 \cos \pi t dt \right]$$

$$= \left(-\frac{n}{\pi} \right) \cdot \left[1 \cdot (-1) - \frac{1}{\pi} \sin \pi t \Big|_0^1 \right] = \frac{n}{\pi}$$

$$\Rightarrow S(n+1) = \frac{n+1}{\pi} \Rightarrow \frac{n}{\pi} \leq S(x) < \frac{n+1}{\pi}$$

$$\Rightarrow \frac{\frac{1}{\pi} \cdot n}{n+1} \leq \frac{S(x)}{x} < \frac{\frac{1}{\pi} \cdot (n+1)}{n}, \text{ 令 } n \rightarrow \infty, \text{ 由夹逼准则: } \lim_{x \rightarrow +\infty} \frac{S(x)}{x} = \frac{1}{\pi}$$

20. (本题 12 分)

设 $f(x)$ 在 $[-1, 1]$ 上二阶连续可导, $f(0)=0$. 证明: 存在 $\xi \in (-1, 1)$, 使得 $f''(\xi) = 3 \int_{-1}^1 f(x) dx$.

解:

$$\text{令 } F(x) = \int_0^x f(t) dt \Rightarrow \int_{-1}^1 f(x) dx = F(1) - F(-1), F(0) = F'(0) = 0.$$

$$F(1) = F(0) + F'(0) \cdot 1 + \frac{F''(0)}{2!} \cdot 1^2 + \frac{F'''(\xi_1)}{3!} \cdot 1^3 \quad \xi_1 \in (0, 1)$$

$$F(-1) = F(0) + F'(0) \cdot (-1) + \frac{F''(0)}{2!} \cdot (-1)^2 + \frac{F'''(\xi_2)}{3!} \cdot (-1)^3 \quad \xi_2 \in (-1, 0)$$

$$F(1) - F(-1) = \frac{F'''(\xi_1) + F'''(\xi_2)}{6} \Rightarrow 3 \int_{-1}^1 f(x) dx = \frac{f'''(\xi_1) + f'''(\xi_2)}{2}$$

$$\text{由介值定理, } \exists \xi \in [\xi_2, \xi_1] \subset (-1, 1), \text{ 使得: } f''(\xi) = \frac{f'''(\xi_1) + f'''(\xi_2)}{2} \Rightarrow 3 \int_{-1}^1 f(x) dx = f''(\xi)$$

21. (本题 13 分)

$$\text{设连续函数 } f(x) \text{ 满足 } \int_0^x f(t) dt + \int_0^x t f(x-t) dt = -\frac{1}{2} e^{-x} \sin x + \frac{1}{2} x.$$

(1) 求 $f(x)$ 的表达式;

(2) 求曲线 $y = f(x)$ ($x \geq 0$) 与 x 轴围成的无界图形的面积.

解: 第一问本质是解微分方程, 但并不代表没学微分方程就没法做! 先通过求导, 去掉变限积分.

$$(1) \text{ 由区间再现, } \int_0^x t f(x-t) dt = \int_0^x (x-t) f(t) dt = x \int_0^x f(t) dt - \int_0^x t f(t) dt.$$

$$\text{故 } \int_0^x f(t) dt + x \int_0^x f(t) dt - \int_0^x t f(t) dt = -\frac{1}{2} e^{-x} \sin x + \frac{1}{2} x, \text{ 两边求导, 得}$$

$$f(x) + \int_0^x f(t) dt + x f(x) - x f(x) = -\frac{1}{2} (-e^{-x} \sin x + e^{-x} \cos x) + \frac{1}{2},$$

$$\text{即 } f(x) + \int_0^x f(t) dt = -\frac{1}{2} (-e^{-x} \sin x + e^{-x} \cos x) + \frac{1}{2},$$

这里可以两边继续求导, 干掉变限积分,

$$\text{但也可两边乘以 } e^x, \text{ 得 } e^x \left[f(x) + \int_0^x f(t) dt \right] = -\frac{1}{2} (-\sin x + \cos x) + \frac{e^x}{2} = \frac{1}{2} (\sin x - \cos x) + \frac{e^x}{2},$$

$$\text{左边显然是 } \left(e^x \int_0^x f(t) dt \right)', \text{ 故在上式两边积分, 得 } e^x \int_0^x f(t) dt = \frac{1}{2} (-\cos x - \sin x) + \frac{e^x}{2} + C,$$



令 $x=0$, 得 $0 = -\frac{1}{2} + \frac{1}{2} + C$, 故 $C=0$, 故 $e^x \int_0^x f(t) dt = \frac{1}{2}(-\cos x - \sin x) + \frac{e^x}{2}$,

两边除以 e^x 并求导, 得 $f(x) = e^{-x} \sin x$.

(2) 由于 $f(x)$ 有正有负, 所以 $S = \int_0^{+\infty} |f(x)| dx = \int_0^{+\infty} e^{-x} |\sin x| dx$,

令 $x + \pi = t$, $S = \int_{\pi}^{+\infty} e^{\pi-t} |\sin(t-\pi)| dt = e^{\pi} \int_{\pi}^{+\infty} e^{-t} |\sin t| dt$

$= e^{\pi} \left[\int_0^{+\infty} e^{-t} |\sin t| dt - \int_0^{\pi} e^{-t} \sin t dt \right] = e^{\pi} \left[S - \int_0^{\pi} e^{-t} \sin t dt \right]$.

令 $J = \int_0^{\pi} e^{-t} \sin t dt$, 连续两次分部积分, 得——

$J = - \int_0^{\pi} \sin t de^{-t} = -\sin t \cdot e^{-t} \Big|_0^{\pi} + \int_0^{\pi} e^{-t} \cos t dt = - \int_0^{\pi} \cos t de^{-t} = -e^{-t} \cos t \Big|_0^{\pi} - \int_0^{\pi} e^{-t} \sin t dt$

$= -(-e^{-\pi} - 1) - J = e^{-\pi} + 1 - J$, 解得 $J = \frac{e^{-\pi} + 1}{2}$.

代入得面积 $S = e^{\pi} \left(S - \int_0^{\pi} e^{-t} \sin t dt \right) = e^{\pi} \left(S - \frac{e^{-\pi} + 1}{2} \right)$, 解得 $S = \frac{-\frac{e^{-\pi} + 1}{2}}{1 - e^{\pi}} = \frac{1}{2} \frac{e^{\pi} + 1}{e^{\pi} - 1}$.

22. (本题 13 分)

设 $f(x)$ 可导, $f(0)=0$, $f'(x)$ 单调减少.

(1) 证明: 当 $x \in (0, 1)$ 时, 有 $f(1)x < f(x) < f'(0)x$;

(2) 若 $f(1) \geq 0$, $f'(0) \leq 1$, 任取 $x_0 \in (0, 1)$, $x_n = f(x_{n-1})$, $n=1, 2, \dots$, 证: $\lim_{n \rightarrow \infty} x_n$ 存在并求其值.

解: (1) 当 $x \in (0, 1)$ 时, 令 $g(x) = f(x) - f'(0)x$, 则

$g'(x) = f'(x) - f'(0) < 0$ (因 $f'(x)$ 单调减少),

故 $g(x)$ 单调减少. 又 $g(0) = 0$, 所以 $g(x) < g(0) = 0$ 即 $f(x) < f'(0)x$.

令 $h(x) = \frac{f(x)}{x}$, 只需证 $\frac{f(1)}{1} < \frac{f(x)}{x}$, 即 $f(1)x < f(x)$,

$h'(x) = \frac{f'(x)x - f(x)}{x^2} = \frac{f'(x)x - [f(x) - f(0)]}{x^2} = \frac{f'(x)x - f'(\xi)x}{x^2} (0 < \xi < x)$,

由 $f'(x)$ 单调减少, 得 $h'(x) < 0$, 所以 $h(x)$ 单调减少,

$h(x) > h(1) = \frac{f(1)}{1}$, 故 $\frac{f(x)}{x} > f(1) \Rightarrow f(1)x < f(x)$

综上, 当 $x \in (0, 1)$ 时, $f(1)x < f(x) < f'(0)x$.

(2) 用数学归纳法证明 $x_n \in (0, 1)$.

当 $n=1$ 时,

$0 \leq f(1)x_0 < x_1 = f(x_0) < f'(0)x_0 \leq x_0 < 1$

假设当 $n=k$ 时, 有 $x_k \in (0, 1)$, 则当 $n=k+1$ 时,

$0 \leq f(1)x_k < x_{k+1} = f(x_k) < f'(0)x_k \leq x_k < 1$ 因此 $x_n \in (0, 1)$.

由 $x_{n+1} = f(x_n) < f'(0)x_n \leq x_n$, 知 $\{x_n\}$ 单调减少, 故 $\lim_{n \rightarrow \infty} x_n$ 存在, 记为 A , 则 $A \geq 0$,

由 $x_n = f(x_{n-1})$ 知 $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} f(x_{n-1})$, 即 $A = f(A)$, 若 $A \in (0, 1)$, 则

$A = f(A) < f'(0)A \leq A$, 矛盾, 故 $A = 0$, 即 $\lim_{n \rightarrow \infty} x_n = 0$.

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